

James Simon

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EDUCATION

University of Southern California B.S. in Computer Science
Dean's List 2023, 2024 | Academic Achievement Award 2024

Expected May 2027

EXPERIENCE

Pixar | AI Systems Intern

06/1/2026–8/21/2026

- Architected a lean knowledge sharing system for Claude agents that was adopted across the company
- Built an agent-driven workflow capturing knowledge gaps and continuously reorganizing documentation
- Researched efficient methods for inference-based file navigation of succinct information
- Developed adaptive online memory-prediction models supporting evolving metadata and historical estimates
- Delivered sub-20-ms single-CPU predictions, reducing MAE 14% below baseline to 1.8GB MAE
- Automated server performance testing with ApacheBench for learning and prediction across Kubernetes pods
- Engineered Parquet data pipelines for 70M+ render-farm tasks for modeling and deployment

OpenAI | AI Fellowship/Contract Work

09/2025–12/2025

- Supported the image generation team's efforts to train the models on real-world user feedback
- Synthesized user telemetry into actionable improvements for model inference
- Engineered data for complex sentiment analysis to strengthen model appraisal and correction systems
- Quantified reproducible success metrics to scale across images of all contexts

INTERDEPENDENT Studios | AI Systems Architect

03/2025–09/2025

- Automated pipeline processing 5,000+ screenplays using Python models & LLM APIs through a web interface
- Built multimodal ML framework supporting 2 multimillion-dollar film greenlighting decisions
- Visualized the screenplay evaluation pipeline granularity to be more approachable or technical
- Engineered an NLP model that estimated the IMDb critic score of 90% of screenplays within 1 point

USC Rocket Propulsion Laboratory | Full-Stack Team Lead

08/2024–Present

- Built mission control web application with real-time telemetry streaming in scalable AWS EC2 instance
- Developed full-stack materials ordering system with React, TypeScript, Express, SQLite
- Containerized applications using Docker and Docker Compose deployed on AWS infrastructure
- Managed in-lab server, sprint planning, office hours, and all third-party services

RESEARCH & PROJECTS

IBM x Q³ Quantum Computing Hackathon | Finalist

September 2025

- Assessed quantum computing viability for PFAS bond dissociation using VQE with UCCSD ansatz (Qiskit)
- Compared quantum versus classical computational methods (CCSD, DFT) and analyzed convergence behavior
- Implemented quantum circuit optimization for compounds that were simulated on IBM quantum computers

USC Brain, Emotion, and Attention Development Lab | Research Assistant

January 2026+

- Rebuilt old PHP/XML app into an offline-capable React/Node.js iPad PWA for an infant neurocognitive study
- Preserved 1 ms-accurate EEG/fNIRS synchronization and developed supporting data collection and analysis
- Managed research databases and telemetry systems capturing trial-level behavioral data from participants

USC Advanced Games Project | Lead Designer, *Cat Counselor*

November 2025+

- Spearheaded project to gamify anxiety therapy worksheets to make exercises more approachable
- Pitched to a prestigious game thesis pipeline, picked as one of three undergraduates out of 100+ applicants
- Coordinates design on a team of 25+ developers with USC Psychology labs and clinical mentors